Integers and Order of Operations

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS integer *(tau tōpū)* · negative *(tōraro)* · positive *(tōrunga)* · order of operations *(raupapa mahi)*

A Adding and subtracting integers ●○○

Remember: Subtracting a negative is the same as adding: $6 - (-2) = 6 + 2$. Adding a negative is the same as subtracting: $6 + (-2) = 6 - 2$.

1. Work out.

a) $-3 + 7$ = _____ **b)** $5 - 9$ = _____ **c)** $-4 - 6$ = _____ **d)** $-8 + 3$ = _____

e) $6 - (-2)$ = _____ **f)** $-5 - (-9)$ = _____ **g)** $-7 + (-4)$ = _____ **h)** $12 - 20$ = _____

i) $-15 + 15$ = _____ **j)** $-2 - 3 + 8$ = _____ **k)** $4 - 10 - (-3)$ = _____ **l)** $-6 + 2 - 5$ = _____

m) $-25 + 40$ = _____ **n)** $13 - (-13)$ = _____ **o)** $-30 - 45$ = _____ **p)** $-1 - (-1)$ = _____

B Multiplying and dividing integers ●○○

Watch out: Two signs the same give a positive answer. Two different signs give a negative answer. Also $(-4)^2 = 16$, but $-4^2 = -16$.

2. Work out.

a) -3×4 = _____ **b)** $-5 \times (-6)$ = _____ **c)** $18 \div (-3)$ = _____ **d)** $-24 \div (-8)$ = _____

e) $7 \times (-2)$ = _____ **f)** $-36 \div 4$ = _____ **g)** $(-4)^2$ = _____ **h)** -4^2 = _____

i) $(-2)^3$ = _____ **j)** -9×0 = _____ **k)** $(-1)^5$ = _____ **l)** $-2 \times 3 \times (-5)$ = _____

m) $-60 \div (-12)$ = _____ **n)** $11 \times (-11)$ = _____ **o)** $-100 \div 25$ = _____ **p)** $(-10)^3$ = _____

C Order of operations ●●○

Remember: GEMA: Grouping (brackets) first, then Exponents, then Multiplication and division, then Addition and subtraction.

3. Work out. Show your first step.

a) $3 + 4 \times 5$ **b)** $20 - 12 \div 4$ **c)** $(3 + 4) \times 5$

d) 2×3^2 **e)** $18 \div (2 + 4)$ **f)** $5 + 2 \times (8 - 3)$

g) $4^2 - 3 \times 5$ **h)** $(6 + 2)^2 \div 4$ **i)** $30 - 2 \times 3 + 4$

j) $7 \times 2 + 6 \div 3$ **k)** $50 - 5^2 \times 2$ **l)** $(9 - 5) \times (2 + 6)$

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D Order of operations with negatives ●●●

4. Work out. Show your first step.

a) $-3 + 4 \times (-2)$ **b)** $(-5)^2 - 10$ **c)** $6 - 2 \times (-4)$

d) $-12 \div (1 - 4)$ **e)** $(-2 + 7) \times (-3)$ **f)** $-4 \times 5 + 30$

g) $2 \times (-3)^2$ **h)** $-20 \div 4 - 6$ **i)** $(3 - 8) \times (2 - 6)$

E Fraction bars ●●●

> **Remember:** A fraction bar works like brackets: work out the top and the bottom first, then divide.

5. Work out.

a) $\dfrac{12 + 8}{4}$ **b)** $\dfrac{30}{2 + 3}$ **c)** $\dfrac{6 \times 4}{10 - 2}$

d) $\dfrac{5^2 - 1}{6}$ **e)** $\dfrac{18 - 6}{-3}$ **f)** $\dfrac{4 + 2 \times 7}{3^2}$

F Problems ●●●

6. Write a calculation for each problem, then work it out.

a) At night the temperature is −4°C. By noon it has risen by 11°C. What is the temperature at noon?

b) A diver is 12 m below the surface of the sea, at -12 m. She swims down another 9 m. What is her new position?

c) A bank account is overdrawn by \$35, so its balance is $-\$35$. Then \$80 is paid in. What is the new balance?

d) The temperature falls from 5°C to −8°C. By how many degrees does it fall?

e) A lift starts on floor 3. It goes down 5 floors and then up 9 floors. Which floor is it on now?

f) In a quiz you get 5 points for a right answer and lose 2 points for a wrong answer. Find the score for 8 right answers and 6 wrong answers.

★ Extension

7. Put one pair of brackets into each calculation to make it true.

a) $3 + 4 \times 5 = 35$ **b)** $20 - 12 \div 4 + 2 = 18$ **c)** $2 \times 5 - 1^2 = 32$

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Integers and Order of Operations

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS integer *(tau tōpū)* · negative *(tōraro)* · positive *(tōrunga)* · order of operations *(raupapa mahi)*

A Adding and subtracting integers ●○○

Remember: Subtracting a negative is the same as adding: $6 - (-2) = 6 + 2$. Adding a negative is the same as subtracting: $6 + (-2) = 6 - 2$.

1. Work out.

a) $-4 + 9$ = _____ **b)** $3 - 8$ = _____ **c)** $-5 - 7$ = _____ **d)** $-9 + 2$ = _____

e) $5 - (-3)$ = _____ **f)** $-4 - (-10)$ = _____ **g)** $-6 + (-5)$ = _____ **h)** $14 - 21$ = _____

i) $-18 + 18$ = _____ **j)** $-3 - 4 + 9$ = _____ **k)** $5 - 12 - (-4)$ = _____ **l)** $-7 + 3 - 6$ = _____

m) $-35 + 50$ = _____ **n)** $16 - (-16)$ = _____ **o)** $-40 - 25$ = _____ **p)** $-2 - (-2)$ = _____

B Multiplying and dividing integers ●○○

Watch out: Two signs the same give a positive answer. Two different signs give a negative answer. Also $(-4)^2 = 16$, but $-4^2 = -16$.

2. Work out.

a) -4×6 = _____ **b)** $-3 \times (-7)$ = _____ **c)** $20 \div (-4)$ = _____ **d)** $-27 \div (-9)$ = _____

e) $8 \times (-3)$ = _____ **f)** $-42 \div 6$ = _____ **g)** $(-5)^2$ = _____ **h)** -5^2 = _____

i) $(-3)^3$ = _____ **j)** -7×0 = _____ **k)** $(-1)^6$ = _____ **l)** $-3 \times 2 \times (-4)$ = _____

m) $-72 \div (-12)$ = _____ **n)** $12 \times (-12)$ = _____ **o)** $-120 \div 30$ = _____ **p)** $(-10)^4$ = _____

C Order of operations ●●○

Remember: GEMA: Grouping (brackets) first, then Exponents, then Multiplication and division, then Addition and subtraction.

3. Work out. Show your first step.

a) $2 + 3 \times 6$ **b)** $30 - 18 \div 6$ **c)** $(2 + 3) \times 6$

d) 3×2^3 **e)** $24 \div (3 + 5)$ **f)** $4 + 3 \times (9 - 5)$

g) $5^2 - 4 \times 3$ **h)** $(5 + 1)^2 \div 9$ **i)** $40 - 3 \times 4 + 5$

j) $6 \times 3 + 8 \div 2$ **k)** $36 - 3^2 \times 4$ **l)** $(8 - 5) \times (3 + 4)$

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D Order of operations with negatives ●●●

4. Work out. Show your first step.

a) $-5 + 3 \times (-2)$ **b)** $(-4)^2 - 20$ **c)** $8 - 3 \times (-2)$

d) $-18 \div (2 - 5)$ **e)** $(-3 + 9) \times (-2)$ **f)** $-6 \times 4 + 40$

g) $3 \times (-2)^2$ **h)** $-30 \div 5 - 4$ **i)** $(2 - 9) \times (1 - 4)$

E Fraction bars ●●●

> **Remember:** A fraction bar works like brackets: work out the top and the bottom first, then divide.

5. Work out.

a) $\dfrac{15 + 9}{6}$ **b)** $\dfrac{42}{4 + 3}$ **c)** $\dfrac{8 \times 3}{9 - 3}$

d) $\dfrac{6^2 - 1}{7}$ **e)** $\dfrac{20 - 4}{-2}$ **f)** $\dfrac{6 + 3 \times 10}{2^2}$

F Problems ●●●

6. Write a calculation for each problem, then work it out.

a) At night the temperature is −6°C. By noon it has risen by 14°C. What is the temperature at noon?

b) A diver is 15 m below the surface of the sea, at -15 m. She swims down another 8 m. What is her new position?

c) A bank account is overdrawn by \$45, so its balance is $-\$45$. Then \$70 is paid in. What is the new balance?

d) The temperature falls from 3°C to −9°C. By how many degrees does it fall?

e) A lift starts on floor 4. It goes down 6 floors and then up 8 floors. Which floor is it on now?

f) In a quiz you get 4 points for a right answer and lose 3 points for a wrong answer. Find the score for 9 right answers and 5 wrong answers.

★ Extension

7. Put one pair of brackets into each calculation to make it true.

a) $2 + 6 \times 4 = 32$ **b)** $30 - 18 \div 6 + 3 = 28$ **c)** $3 \times 4 - 2^2 = 12$

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